Chemistry Topic 4 — Acids, Bases, Salts and Carbohydrates

Covers syllabus (click to expand)

Acid-Base Theories (Arrhenius, Brønsted-Lowry, Lewis) | pH Scale & Mathematical Dynamics | Universal, Synthetic & Natural Indicators (Litmus, Phenolphthalein, Methyl Orange, Turmeric) | Natural Organic Acids & Sources (Acetic, Formic, Citric, Tartaric, Lactic, Oxalic, Malic) | Neutralization & Buffer Systems (Human Blood pH Control) | Important Industrial & Household Salts (Baking Soda, Washing Soda, Bleaching Powder, Plaster of Paris, Gypsum, Vitriols, Potash Alum) | Carbohydrates & Sugar Chemistry (Glucose, Fructose, Sucrose, Lactose, Maltose, Invert Sugar, Artificial Sweeteners)

Sources baked in: NCERT Class 10 (Acids, Bases and Salts), Class 11 & 12 (Equilibrium; Biomolecules), Lucent General Science (Chemistry), Physics Wallah (PW) State PCS Foundation, Ghatna Chakra General Science (1990–2025 PYQs). **Weight:** ★★★ High — Recurring questions on baking soda vs washing soda formulas, Plaster of Paris vs gypsum water of crystallization, natural acid-source matches (oxalic in tomatoes, methanoic in ant stings), litmus color transitions, blood pH buffer range, and carbohydrate sweetness hierarchy. **Last verified:** September 2026

Current Affairs (2024–2026)

Year	Event	Core Concept & Must-Score Significance
2025–2026	**Global Ocean Acidification Monitoring**	Ocean absorption of anthropogenic CO_2 produces carbonic acid ($CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$), dropping surface ocean pH from pre-industrial 8.25 to ≈ 8.14. Even this 0.1 unit drop represents a $\approx 30\%$ increase in $[H^+]$ activity, severely impairing calcium carbonate ($CaCO_3$) calcification by coral reefs and molluscs.
2025	**UPPCS Prelims 2025**	Direct question on **pH Scale Inversion & Acid-Base Identification**: Keyed that statements claiming "Acids have pH > 7" or "Bases have pH < 7" are unequivocally false. Re-tested baking soda ($NaHCO_3$) decomposition in soda-acid fire extinguishers ($2NaHCO_3 + H_2SO_4 \to Na_2SO_4 + 2H_2O + 2CO_2 \uparrow$).
2024	**UPPCS Prelims 2024**	Question on **Litmus Paper Response**: Acid turns blue litmus paper red; base turns red litmus blue. Litmus is derived naturally from **Lichens** (symbiotic association of fungus and alga).
2024	**UPPCS / UKPCS Trends**	Formula: **Baking Soda is $NaHCO_3$** (Sodium hydrogen carbonate); **Washing Soda is $Na_2CO_3 \cdot 10H_2O$** (Sodium carbonate decahydrate); **Plaster of Paris is $CaSO_4 \cdot \frac{1}{2}H_2O$**; **Gypsum is $CaSO_4 \cdot 2H_2O$**.

Year	Event	Core Concept & Must-Score Significance
2023	**Biomolecules & Sweeteners**	Testing of **Relative Sweetness Index**: **Fructose** is the sweetest naturally occurring sugar (1.7\times sweeter than sucrose). Cane sugar (**Sucrose**) is a non-reducing disaccharide composed of \alpha\text{-D-glucose} + \beta\text{-D-fructose}, hydrolyzing to yield optically inverted **Invert Sugar**.

📋 **Master Reference Tables (Click to Expand)**

Table 1: Master pH Spectrum of Common Substances (UPPCS Hierarchy) | Substance / Biological Fluid | Exact pH Value / Typical Range | Acidic / Neutral / Basic | Key Point & Physiological Significance | |---|---|---|---| | **Concentrated Battery Acid (H_2SO_4)** | < 1.0 (\sim 0.5) | Strongly Acidic | Electrolyte in automotive lead-acid storage batteries | | **Human Gastric Juice (HCl)** | **1.0–1.5** | Strongly Acidic | Activates pepsinogen; destroys ingested microorganisms | | **Lemon Juice** | **2.2–2.4** | Strongly Acidic | Rich in citric acid (C_6H_8O_7) | | **Vinegar (Sirka)** | **2.4–3.0** | Moderately Acidic | 4–8\% aqueous solution of acetic acid (CH_3COOH) | | **Wine / Aerated Soda Drinks** | 3.0–3.8 | Moderately Acidic | Dissolved carbonic acid (H_2CO_3) and organic fruit acids | | **Tomato Juice** | **4.0–4.4** | Moderately Acidic | Rich in oxalic and citric acids | | **Black Coffee** | 4.8–5.2 | Weakly Acidic | Chlorogenic and quinic acids | | **Acid Rain (Threshold)** | **< 5.6** | Acidic | Caused by atmospheric SO_2 and NO_x dissolving into rain | | **Cow's Fresh Milk** | **6.4–6.6** | Slightly Acidic | Contains lactic acid precursor lactose and casein micelles | | **Human Saliva (Resting)** | **6.5–7.5** | Near Neutral | Demineralization of tooth enamel triggers if \text{pH} < 5.5 | | **Pure Distilled Water (25^\circ\text{C})** | **7.0** | **Strictly Neutral** | [H^+] = [OH^-] = 10^{-7}\text{ M} | | **Human Tears** | **7.4** | Slightly Basic | Contains lysozyme enzyme to lyse bacterial cell walls | | **Human Blood** | **7.35–7.45** | **Slightly Basic** | Strictly buffered by H_2CO_3/HCO_3^-; outside 7.0–7.8 is fatal | | **Sea Water** | **8.0–8.5** | Moderately Basic | Buffered by dissolved bicarbonates, borates, and carbonates | | **Baking Soda (NaHCO_3) Solution** | **8.3–8.5** | Basic / Alkaline | Hydrolysis of weak acid anion (HCO_3^-) yields mild base | | **Washing Soda (Na_2CO_3) Solution** | **10.5–11.0** | Strongly Basic | High alkalinity saponifies grease in laundry washing | | **Milk of Magnesia (Mg(OH)_2)** | **10.5** | Strongly Basic | Standard household antacid suspension and laxative | | **Household Ammonia (NH_4OH)** | 11.5 | Strongly Basic | Glass and window surface cleaner | | **Household Bleach (NaOCl / CaOCl_2)**| 12.5 | Strongly Basic | Disinfectant, hypochlorite oxidizing agent | | **1\text{ M } NaOH (Caustic Soda)** | **14.0** | Extremely Basic | Completely dissociated strong alkali ([OH^-] = 1.0\text{ M}) | --- ### Table 2: Natural Organic Acids & Their Diagnostic Occurrence | Organic Acid | Systematic Chemical Formula | Diagnostic Natural / Household Sources | Special Facts & Applications | |---|---|---|---| | **Acetic Acid** (Ethanoic) | CH_3COOH | **Vinegar** (4–8\% solution) | Food preservative, pickling; 100\% anhydrous form is "Glacial Acetic Acid" (freezes at 16.6^\circ\text{C}). | | **Formic Acid** (Methanoic) | HCOOH | **Ant stings**, bee stings, stinging nettle hairs | Simplest carboxylic acid; neutralized on skin by applying mild alkaline baking soda or calamine (ZnCO_3). | | **Citric Acid** | C_6H_8O_7 | **Lemons, oranges, limes**, grapefruits | Tricarboxylic acid; food acidulant, descaling agent, Krebs cycle metabolite. | | **Tartaric Acid** | C_4H_6O_6 | **Tamarind (imli)**, unripe grapes, cream of tartar | Dicarboxylic dihydroxy acid; essential neutralizer in **baking powder** to prevent alkaline bitterness. | | **Lactic Acid** | CH_3CH(OH)COOH | **Sour milk, curd (dahi)**, yogurt | Produced by *Lactobacillus* fermentation; anaerobic muscular buildup causes exercise cramps. | |

Oxalic Acid | (COOH)_2 \text{ or } H_2C_2O_4 | **Tomatoes, spinach**, sorrel, rhubarb | Forms insoluble **Calcium Oxalate (CaC_2O_4) kidney stones**; removes rust and ink stains from clothing. | | **Malic Acid** | C_4H_6O_5 | **Apples** (especially green apples), pears | Dicarboxylic acid responsible for sour taste of unripe deciduous pome fruits. | | **Ascorbic Acid** | C_6H_8O_6 | **Amla**, guava, green peppers, citrus | **Vitamin C**; water-soluble antioxidant; deficiency causes scurvy (bleeding gums). | | **Tannic Acid** | Complex polyphenol | **Tea leaves**, oak bark, gallnuts | Astringent taste of over-steeped tea; used in leather tanning and iron-gall ink. | | **Butyric Acid** | C_3H_7COOH | **Rancid butter**, aged parmesan cheese | Foul, vomit-like odor generated by triglyceride hydrolysis by lipase enzymes. | --- ### Table 3: Master Classification of Industrial & Medicinal Salts | Commercial Name | Systematic Chemical Name | Exact Chemical Formula | Industrial Preparation Method | Primary Uses & Must-Score | |---|---|---|---|---| | **Baking Soda** | Sodium Hydrogen Carbonate | **NaHCO_3** | Solvay Process (NaCl + H_2O + NH_3 + CO_2 \to NaHCO_3 + NH_4Cl) | Antacid, baking powder ingredient, soda-acid fire extinguishers. | | **Washing Soda** | Sodium Carbonate Decahydrate | **Na_2CO_3 \cdot 10H_2O** | Recrystallization of Solvay soda ash | Laundry detergent, glass/paper mills, removes permanent water hardness. | | **Bleaching Powder** | Calcium Oxychloride | **CaOCl_2** | Chlorination of dry slaked lime (Ca(OH)_2 + Cl_2 \xrightarrow{40^\circ\text{C}} CaOCl_2 + H_2O) | Sterilizing municipal drinking water, textile bleaching, oxidizer. | | **Plaster of Paris** | Calcium Sulphate Hemihydrate | **CaSO_4 \cdot \frac{1}{2}H_2O** | Heating gypsum to 373\text{ K} (100^\circ\text{C}) in a rotary kiln | Orthopedic fracture casts, false ceilings, statues; expands \approx 1\% on setting. | | **Gypsum** | Calcium Sulphate Dihydrate | **CaSO_4 \cdot 2H_2O** | Natural mineral extraction | Added (2–3\%) to cement to retard flash setting; alkali soil reclamation. | | **Blue Vitriol** | Copper Sulphate Pentahydrate | **CuSO_4 \cdot 5H_2O** | Reaction of copper oxide/carbonate with H_2SO_4 | Bordeaux mixture agricultural fungicide, electroplating, battery electrolyte. | | **Green Vitriol** | Ferrous Sulphate Heptahydrate | **FeSO_4 \cdot 7H_2O** | Scrap iron reacting with dilute H_2SO_4 | Iron anaemia medicine, black ink manufacture, sewage coagulant. | | **White Vitriol** | Zinc Sulphate Heptahydrate | **ZnSO_4 \cdot 7H_2O** | Reaction of zinc or ZnO with dilute H_2SO_4 | Lithopone pigment, zinc dietary supplements, eye drops. | | **Potash Alum** | Potassium Aluminium Sulphate | **K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O**| Double crystallization of equimolar K_2SO_4 and Al_2(SO_4)_3 | Water purification coagulant, hemostatic styptic (stops cut bleeding). | | **Epsom Salt** | Magnesium Sulphate Heptahydrate | **MgSO_4 \cdot 7H_2O** | Mining of kieserite mineral | Saline osmotic purgative laxative, therapeutic mineral bath soak. |

Detailed Theory Notes

4.1 Acid-Base Theories & Mathematical Dynamics of the pH Scale

1. Theoretical Evolution

- **Arrhenius Concept (1884)**: Defined acids as substances producing H^+ ions in aqueous solution, and bases as producing OH^- ions. Limitation: Restricted strictly to aqueous systems and failed to explain the basicity of ammonia (NH_3).
- **Brønsted-Lowry Concept (1923)**: Acid is a proton (H^+) donor; base is a proton (H^+) acceptor. Introduces **Conjugate Acid-Base Pairs** differing by exactly one proton (H^+):

\underset{\text{Acid}_1}{HCl} + \underset{\text{Base}_2}{H_2O} \rightleftharpoons \underset{\text{Acid}_2}{H_3O^+} + \underset{\text{Base}_1}{Cl^-}

- **Lewis Concept (1923)**: Broadest electronic definition:
- *Lewis Acid*: Chemical species capable of accepting an unshared electron pair (electrophile, e.g., BF_3, AlCl_3, Fe^{3+}, H^+).
- *Lewis Base*: Chemical species capable of donating an unshared electron pair (nucleophile, e.g., :NH_3, H_2\ddot{O}:, OH^-, CN^-).

```
Arrhenius:      Acid = H⁺ generator in water     | Base = OH⁻ generator in water
Brønsted-Lowry: Acid = Proton (H⁺) donor         | Base = Proton (H⁺) acceptor
Lewis:          Acid = Electron pair acceptor    | Base = Electron pair donor
```

2. Logarithmic pH Scale Dynamics

- Formulated mathematically by S.P.L. Sørensen (1909): \text{pH} = -\log_{10}[H_3O^+]
- Because the scale is base-10 logarithmic, a decrease of 1 unit on the pH scale represents a **10-fold increase in hydrogen ion concentration**: \frac{[H^+]*{\text{pH } 2}}{[H^+]} 5}} = \frac{10^{-2}}{10^{-5}} = 10^3 = 1000\times \text{ higher acidity
- **Neutrality Definition**: Neutrality does NOT universally mean \text{pH} = 7.0; neutrality strictly means [H^+] = [OH^-]. At 25^\circ\text{C}, K_w = 1.0 \times 10^{-14}, so \text{pH} = 7.0. At elevated physiological temperatures (37^\circ\text{C}), K_w \approx 2.4 \times 10^{-14}, so neutral \text{pH} \approx 6.8.

4.2 Chemical Indicators & Titration Physics

1. Action of Acid-Base Indicators

Indicators are weak organic acids (HIn) or weak organic bases whose undissociated molecular form has a visibly distinct color from its ionized conjugate form (In^-): HIn\text{ (Color A)} \rightleftharpoons H^+\text{(aq)} + In^-\text{ (Color B)} - In an acidic solution (high [H^+]), common-ion effect suppresses indicator dissociation, displaying Color A. - In an alkaline solution, added OH^- consumes H^+, shifting the equilibrium forward to display Color B.

2. Master Indicator Spectrum Table

Indicator Name	Nature / Source	Acidic Medium Color	Neutral Medium Color	Basic/Alkaline Medium Color	pH Transition Interval
Litmus	Natural (Lichen extract)	**Red / Pink**	Purple	**Blue**	5.0–8.0
Phenolphthalein	Synthetic (Phthalic anhydride + phenol)	**Colorless**	Colorless	**Magenta / Deep Pink**	8.2–10.0
Methyl Orange	Synthetic (Sodium sulfonate azo dye)	**Red / Orange-Red**	Orange	**Yellow**	3.1–4.4
Methyl Red	Synthetic azo dye	**Red**	Orange	**Yellow**	4.4–6.2

Indicator Name	Nature / Source	Acidic Medium Color	Neutral Medium Color	Basic/Alkaline Medium Color	pH Transition Interval
Turmeric (Haldi)	Natural rhizome (*Curcuma longa*)	**Yellow**	Yellow	**Reddish-Brown**	7.4–9.2
Red Cabbage	Natural anthocyanin pigment	**Red / Pink**	Purple	**Green to Yellow**	2.0–12.0

4.3 Household Salts & Industrial Chemistry

1. Baking Soda ($NaHCO_3$) vs. Washing Soda ($Na_2CO_3 \cdot 10H_2O$)

- **Solvay Industrial Process**: $NaCl + H_2O + NH_3 + CO_2 \to NaHCO_3\downarrow + NH_4Cl$ Crude sodium hydrogen carbonate precipitates out because of its relatively low solubility in ammonium chloride brine.
- **Thermal Decomposition into Soda Ash**: $2NaHCO_3 \xrightarrow{\Delta} Na_2CO_3 + H_2O + CO_2 \uparrow$
- **Hydration into Washing Soda**: $Na_2CO_3 + 10H_2O \to Na_2CO_3 \cdot 10H_2O\text{ (Washing Soda Decahydrate)}$
- **Efflorescence Reaction**: In dry atmospheric conditions, washing soda loses 9 molecules of water spontaneously: $Na_2CO_3 \cdot 10H_2O \xrightarrow{\text{dry air}} Na_2CO_3 \cdot H_2O\text{ (Monohydrate)} + 9H_2O \uparrow$

2. Plaster of Paris ($CaSO_4 \cdot \frac{1}{2}H_2O$) vs. Gypsum ($CaSO_4 \cdot 2H_2O$)

- **Thermal Regimes of Gypsum Calcination**: 1. *At 100°C (373 K)*: $CaSO_4 \cdot 2H_2O \xrightarrow{373\text{ K}} CaSO_4 \cdot \frac{1}{2}H_2O\text{ (Plaster of Paris)} + 1\frac{1}{2}H_2O$ 2. *Above 200°C (473 K)*: $CaSO_4 \cdot 2H_2O \xrightarrow{>473\text{ K}} CaSO_4\text{ (Anhydrous "Dead Burnt Plaster")} + 2H_2O$ Dead burnt plaster absorbs water very slowly and has no hydraulic setting strength. 3. *At $>1000^\circ\text{C}$*: $2CaSO_4 \xrightarrow{>1000^\circ\text{C}} 2CaO + 2SO_2 \uparrow + O_2 \uparrow$

4.4 Carbohydrates, Sugars & Sweetener Dynamics

1. Structural Classification

- **Monosaccharides**:
- *D-Glucose*: Aldohexose; exists predominantly as a 6-membered pyranose ring ($\alpha\text{-D-glucopyranose}$ and $\beta\text{-D-glucopyranose}$).
- *D-Fructose*: Ketohexose; exists predominantly as a 5-membered furanose ring ($\alpha\text{-D-fructofuranose}$ and $\beta\text{-D-fructofuranose}$).
- **Disaccharides**:
- *Sucrose*: Composed of $\alpha\text{-D-glucose} + \beta\text{-D-fructose}$ linked by an $\alpha\text{-}1,\beta\text{-}2$ glycosidic bond.

- *Lactose*: Composed of \beta\text{-D-galactose} + \beta\text{-D-glucose} linked by a \beta-1,4 glycosidic bond.
- *Maltose*: Composed of two \alpha\text{-D-glucose} molecules linked by an \alpha-1,4 glycosidic bond.

2. Inversion of Cane Sugar (Invert Sugar)

Sucrose is naturally dextrorotatory ([\alpha]*D = +66.5^\circ). When cleaved by dilute acid or the enzyme invertase: C* Because the levorotation of fructose exceeds the dextrorotation of glucose, the optical rotation of the resulting equimolar mixture is levorotatory (}H_{22}O_{11} + H_2O \xrightarrow{\text{invertase}} \underset{\text{Dextrorotatory } (+52.7^\circ)}{\text{D-Glucose}} + \underset{\text{Strongly Levorotatory } (-92.4^\circ)}{\text{D-Fructose}}**-19.8^\circ**). This inversion of the optical rotation gives **Invert Sugar** its name. Invert sugar is sweeter and more resistant to crystallization than sucrose, making it valued in confectionery and commercial honey.

Bilingual Terminology -- Acids, Bases, Salts and Sucrose

English Term	Hindi Term	Key Anchor
Acid (अम्ल)	अम्ल	Donates H+ (proton); pH < 7; sour taste; turns blue litmus red
Base (क्षार)	क्षार	Accepts H+ (proton); pH > 7; bitter taste, soapy feel; turns red litmus blue
Alkali (क्षारक)	क्षारक	Water-soluble base; NaOH, KOH, Ca(OH)2
pH (पीएच)	pH / पीएच	Negative log of H+ concentration; 0-14 scale; 7 = neutral
Neutralisation (उदासीनीकरण)	उदासीनीकरण	Acid + Base -> Salt + Water; exothermic
Salt (लवण)	लवण	Product of acid-base reaction; e.g., NaCl, CaCO3, CuSO4
Indicator (सूचक)	सूचक	Substance that changes colour with pH; litmus, phenolphthalein, methyl orange
Buffer (बफर)	बफर	Resists change in pH; e.g., blood pH 7.4 maintained by H2CO3/HCO3-
Bleaching Powder (विरंजक चूर्ण)	विरंजक चूर्ण / ब्लीचिंग पाउडर	Ca(OCl)Cl; used for bleaching, disinfecting water
Plaster of Paris (प्लास्टर ऑफ पेरिस)	पेरिस प्लास्टर	CaSO4.1/2H2O; sets hard by absorbing water; used in fractures, moulds
Washing Soda (धावन सोडा)	धावन सोडा	Na2CO3.10H2O; removes permanent hardness; used in glass, soap making
Baking Soda (बेकिंग सोडा)	मीठा सोडा / खाने का सोडा	NaHCO3; releases CO2 on heating; used in baking, antacids
Caustic Soda (दाहक सोडा)	दाहक सोडा / कास्टिक सोडा	NaOH; strong base; soap/paper making; drain cleaner

English Term	Hindi Term	Key Anchor
Vinegar (सिरका)	सिरका	Acetic acid solution (4-8% CH3COOH); sour taste
Hydrochloric Acid (हाइड्रोक्लोरिक अम्ल)	हाइड्रोक्लोरिक अम्ल	HCl; "muriatic acid"; in gastric juice; strong acid
Sulphuric Acid (सल्फ्यूरिक अम्ल)	गंधक का तेजाब	H2SO4; "king of chemicals"; strong acid; highly corrosive

One-Liner Revision -- Acids, Bases, Salts

#	Fast Fact
1	pH scale = 0-14; pH < 7 = acid; pH 7 = neutral; pH > 7 = base/alkaline
2	Strongest acid = HClO4 (perchloric); in chemistry; practically H2SO4 is "king"
3	Common acid in stomach (gastric juice) = **HCl** (hydrochloric acid)
4	Formic acid (HCOOH) = in ant sting; methanoic acid; simplest organic acid
5	Acetic acid (CH3COOH) = vinegar; ethanoic acid
6	Citric acid = lemons, oranges; ascorbic acid = Vitamin C
7	Lactic acid = sour milk (curd); produced by Lactobacillus
8	Oxalic acid = in spinach/tomato leaves; kidney stone-causing
9	Sulphuric acid (H2SO4) = "King of Chemicals"; most produced industrial chemical
10	Aqua Regia = 3 parts HCl + 1 part HNO3; dissolves gold and platinum
11	Litmus = turns red in acid, blue in base; from lichen (Rocella tinctoria)
12	Phenolphthalein = colourless in acid; pink/red in base
13	Methyl orange = red in acid; yellow in base
14	NaOH (Caustic soda) = used in soap-making, paper, textile
15	Ca(OH)2 (slaked lime / calcium hydroxide) = whitewash; water treatment; mortar
16	Baking soda (NaHCO3) = antacid for acidity; releases CO2 in baking
17	Washing soda (Na2CO3.10H2O) = removes permanent hardness; laundry
18	Plaster of Paris (CaSO4.1/2H2O) = made from gypsum (CaSO4.2H2O) by heating
19	Bleaching powder Ca(OCl)Cl = disinfects water; bleaches cloth/paper
20	Common salt (NaCl) = table salt; from sea water by evaporation or rock salt mines
21	Blood pH = 7.4 (slightly basic); maintained by bicarbonate buffer
22	Acid rain pH < 5.6 (H2SO4 + HNO3 from SO2 + NOx emissions)

#	Fast Fact
23	Antacids = neutralise excess HCl in stomach; Mg(OH)2, Al(OH)3, NaHCO3
24	Hard water + soap = scum (Ca/Mg stearate); soft water lathers well
25	Soap is sodium salt of fatty acid; detergent = synthetic sulphonate/sulphate
26	Saponification = hydrolysis of fats/oils with NaOH/KOH to make soap + glycerol
27	Sugar (sucrose) = C12H22O11; invert sugar = glucose + fructose (after acid hydrolysis)
28	Glucose = C6H12O6; dextrose; simplest monosaccharide; brain's primary fuel
29	Fructose = fruit sugar; sweetest naturally occurring sugar
30	Lactose = milk sugar; disaccharide (glucose + galactose); lactose intolerance = deficiency of lactase